Tie Rod

The **tie rod** is the internal threaded rod that draws the Slocum hull sections together and holds the glider closed. It is mounted to the bottom of the aft electronics tray and runs forward through each section, threading into a **tie rod receiver** in the forward section (or in an add-on module). Tightening the rod is what pulls the mating hulls over their O-ring seals and squares them up.

Source

Paraphrased from the *Slocum G3 Glider Maintenance Manual* (Rev. A), the UG2 community Slack, and the Teledyne Webb Research user forum. This is a condensed field reference — always defer to the official Teledyne documentation for your specific glider.

How it works

You reach the tie rod through the **vacuum seal port** on the aft endcap. With the PEEK vacuum seal plug removed, a hex bit driven by a **15 inch-pound torque T-handle** engages the rod end:

Action	Tool	Bit
Main aft tie rod	15 in-lb torque T-handle + 24" extension	3/16" hex
Add-on / energy-bay tie rod extension	15 in-lb torque T-handle	5/32" hex

- **Opening:** unscrew the tie rod until tension releases, then separate the sections with the hull separator tool. The rod must be fully disengaged from its receiver before the sections will come apart.
- **Closing:** route the rod through the **payload bay tie rod guide tube**, bring the sections together, then screw the rod into the forward section (or add-on module) receiver and tighten to **15 inch-pounds**. The sections should draw together smoothly, tight and square, with index marks aligned and O-rings unpinched.